

		$4\pi\epsilon_0 \left(\frac{1}{3}r\right)^2 \quad 4\pi\epsilon_0 \left(\frac{2}{3}r\right)^2 \quad 8\pi\epsilon_0 r^2$
21	D	Using Fleming's Left Hand Rule, the magnetic force on the protons is upwards. Hence, the electric force needs to be applied such that there is a force on the proton downwards so that the protons will not be deflected. Since protons are positive, the direction of the E-field should be downwards, towards the lower plate. $Bqv = qE \rightarrow E = Bv = (1.5)(2 \times 10^7) = 3 \times 10^7 \text{ N C}^{-1}$
22	A	$E = W/Q = (2500+1500) / 2000 = 2 \text{ V}$